

Mathematical Review: National Channel Territory Design and Staffing

An Analytical Assessment of Multi-Scale Formulations, Support-Based Exact Partitioning, Scale Invariance, and Axiomatic Feasible Envy-Freeness

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Abstract

This review presents an exhaustive mathematical and axiomatic appraisal of the multi-channel sales territory design and wholesaler staffing optimization architecture documented across `docs/MODEL_FULL.md`, the operational solvers, and the latest production pipelines. We analyze the historical decomposition spanning state-level mixed-integer linear programming (Level 0), semi-discrete optimal transport with discrete graph repair heuristics (Level 2), and Nash social welfare bipartite matching (Stage 2), detailing the mathematical root causes that produced combinatorial timeouts and heuristic boundary slacks.

To overcome these structural limitations, we formulate and evaluate the newly implemented **Support-Based Exact Reformulation Framework**. By pre-enumerating connected graph supports of bounded spatial diameter (≤ 900 km) on the planning graph and solving an exact binary Master MILP over continuous contact shares, this paradigm guarantees 100% rook contiguity, exact capacity band satisfaction, and zero unheld opportunity mass by construction, solving to provable global optimality in under 3 seconds. Furthermore, we examine the **Unified Multi-Channel Architecture**, which resolves the sparse geography paradox across 7 rural Western states through a dedicated regional merged channel (WIFI/WHFI), realizing 36 certified Pareto-optimal scenarios ($K \in [46, 53]$ total districts).

On the axiomatic front, we prove the exact scale invariance of Nash Social Welfare under multiplicative descaling and demonstrate why multiplicative normalization fails to alter the dimensionless curvature of the welfare function. We show that while a zero disagreement point ($d = \mathbf{0}$) historically anchored the classical Envy-Free up to One Item (EF1) guarantee of Caragiannis et al. (2019), that guarantee is already invalidated by hard capacity bands and spatial contiguity. We formalize **Feasible Envy-Freeness (FEF_x)** under Generalized Assignment Constraints (Barman et al., 2023) paired with asymmetric claims-scaled reservation floors ($d_i = \alpha_i S_i + d_{\text{floor}}$), restoring steep concavity to protect incumbent wholesaler books while guaranteeing bargaining set non-emptiness. Finally, we document the planar polygon dissolution pipeline and the complete 33,791-ZCTA nationwide crosswalk (6,082,380 rows) that eliminates spatial omissions in downstream enterprise reporting.

1 Introduction and Problem Formulation

The commercial sales territory design and staffing problem formulated in `docs/MODEL_FULL.md` addresses the simultaneous partitioning of nationwide sales opportunity across the 48 contiguous United States and Washington, D.C. ($S = 49$ planning states) into K balanced sales districts. Concurrently, it models the assignment of incumbent sales representatives across four fine sales channels:

$$C = \{N_{\text{WH}}, N_{\text{FI}}, \text{WH}, \text{FI}\} \tag{1}$$

admitting seven permissible channel bundle configurations:

$$\mathfrak{B} = \{N, \text{WH}, \text{FI}, \text{WH_PLUS}, \text{FI_PLUS}, \text{WHFI}, \text{WHFI_PLUS}\} \quad (2)$$

where $N = \{N_{\text{WH}}, N_{\text{FI}}\}$, $\text{WH} = \{\text{WH}\}$, $\text{FI} = \{\text{FI}\}$, and $\text{WHFI} = \{\text{WH}, \text{FI}\}$.

1.1 Monolithic MINLP Formulation

Formulated as a monolithic optimization problem over the set of discrete geographic units Z ($|Z| = 6,459$ commercial ZIP codes, or $|Z| = 33,791$ total Census ZCTAs), the mathematical program is an intractable Mixed-Integer Non-Convex Non-Linear Program (MINLP):

$$\max_{x,y,w} \quad \sum_{i \in R} \log(g_i(x, y, w_i)) - \rho \sum_{j=1}^K \text{Perimeter}(A_j) \quad (3)$$

$$\text{s.t.} \quad \sum_{j=1}^K x_{z,j,c} = 1 \quad \forall z \in Z, c \in C \quad (4)$$

$$L_{B_j} \leq \sum_{z \in Z} \sum_{c \in B_j} M_{z,c} x_{z,j,c} \leq U_{B_j} \quad \forall j \in \{1, \dots, K\} \quad (5)$$

$$G[A_j] \text{ is a connected planar subgraph} \quad \forall j \in \{1, \dots, K\} \quad (6)$$

$$\sum_{i \in R} w_{i,j} = 1, \quad \sum_{j=1}^K w_{i,j} \leq 1 \quad \forall j, i \quad (7)$$

$$x_{z,j,c} \in \{0, 1\}, \quad w_{i,j} \in \{0, 1\} \quad \forall z, j, c, i \quad (8)$$

The combination of NP-hard planar graph partitioning, non-linear logarithmic Nash social welfare $\sum \log g_i$, channel bundling product-form constraints, and quadratic perimeter penalties renders direct global optimization across 6,459 geographic units computationally intractable.

To achieve tractability, the operational architecture initially adopted a **three-level hierarchical decomposition**:

1. **Level 0 (Macro Planning)**: A state-by-channel MILP solved sequentially on the state rook graph $G_S = (S, E_S)$.
2. **Level 2 (Micro Realization)**: A semi-discrete optimal transport LP per split state, followed by discrete graph repair heuristics on the Voronoi dual graph G_{cell} .
3. **Stage 2 (Wholesaler Staffing)**: A maximum-weight bipartite matching on logarithmic gains solved via the Kuhn-Munkres (Hungarian) algorithm.

2 The Legacy Decomposition and Its Structural Failure Modes

2.1 Level 0: Macro-Level Channel Planning (State $\times$ Channel MILP)

Level 0 abstracts geography to the state grain. For each slot $j \in J$ belonging to bundle B_j , the decision space comprises:

- $y_{s,j} \in [0, 1]$: Continuous share of state s allocated to slot j .
- $z_{s,j} \in \{0, 1\}$: Binary contact indicator ($y_{s,j} > 0$).
- $u_j \in \{0, 1\}$: Binary slot utilization indicator.
- $r_{s,j} \in \{0, 1\}$: Contiguity tree root indicator.

- $f_{a,j} \in [0, N_S - 1]$: Commodity flow on directed arc $a \in A_S$ where $N_S = |S| = 49$.

The core relations governing Level 0 are:

$$\text{Product-form Coverage: } \sum_{j:c \in B_j} y_{s,j} \leq 1 - p_{s,c} \quad \forall s \in S, c \in C \quad (9)$$

$$\text{Contact Sizing: } \eta z_{s,j} \leq y_{s,j} \leq z_{s,j} \leq u_j \quad \forall s \in S, j \in J \quad (10)$$

$$\text{Capacity Bands: } L_{B_j} u_j \leq \sum_{s \in S} W_{s,j} y_{s,j} \leq U_{B_j} u_j + \sum_{s \text{ capped}} a_s z_{s,j} \quad \forall j \in J \quad (11)$$

$$\text{Tree Contiguity: } \sum_{s \in S} r_{s,j} = u_j, \quad r_{s,j} \leq z_{s,j} \quad \forall j \in J \quad (12)$$

$$f_{a,j} \leq (N_S - 1) z_{\text{tail}(a),j}, \quad f_{a,j} \leq (N_S - 1) z_{\text{head}(a),j} \quad \forall a \in A_S, j \in J \quad (13)$$

$$z_{s,j} - N_S r_{s,j} \leq \sum_{a \in \delta^-(s)} f_{a,j} - \sum_{a \in \delta^+(s)} f_{a,j} \quad \forall s \in S, j \in J \quad (14)$$

where $W_{s,j} = \sum_{c \in B_j} M_{s,c}$, $p_{s,c}$ represents prior committed coverage, and $a_s = \max(0, W_s/\text{cap}_s - \tau_B)$ provides state-specific band-break allowances under split caps cap_s .

2.2 Level 2: Micro-Level Realization (Optimal Transport and Graph Repair)

Given state shares $y_{s,j}$, unsplit states are mapped entirely to district j . Split states are partitioned via the semi-discrete optimal transport LP:

$$\min_x \sum_{z \in s} \sum_j M_z (\|p_z - c_j\|^2 + \pi_{z,j}) x_{z,j} \quad (15)$$

$$\text{s.t. } \sum_j x_{z,j} = 1 \quad \forall z \in s, \quad \sum_{z \in s} M_z x_{z,j} = y_{s,j} M_s \quad \forall j \quad (16)$$

Centroids c_j are iteratively shifted via up to five Lloyd updates. Resulting disconnected components are reconnected post-LP via heuristic frontier expansion and bridging on the Voronoi dual graph G_{cell} with capacity slack $\sigma = 0.02\tau_B$.

2.3 Structural Breakdown and Failure Modes of the Legacy Pipeline

Rigorous mathematical auditing identified four critical vulnerabilities in this decomposition:

1. **Combinatorial Explosion in Multi-Channel MILPs:** When moving from single-channel National (14 districts) to multi-channel execution (WH with 11–14 districts and FI with 21–26 districts), the binary flow-tree formulation (12)–(14) suffered catastrophic branch-and-bound growth. Solvers frequently exceeded 600-second timeouts without closing optimality gaps.
2. **Topological Disconnection under Convex Power Cells:** By the Brenier-Aurenhammer theorem of optimal transport, the continuous LP solutions partition Euclidean space into convex polyhedral power diagrams. When intersecting convex power cells with non-convex administrative geographies (coastal inlets, peninsulas, river boundaries, panhandles), discrete ZIP code assignments frequently fragmented into disconnected topological islands.

3. **Heuristic Slippage and Unheld Mass:** The post-hoc discrete repair heuristics on G_{cell} lacked global guarantees. To bridge disconnected islands across state lines, the heuristic permitted $\pm 2\%$ capacity slacks, frequently causing unheld mass ($> 0.5\%$) or subtle rook-contiguity violations in multi-county corridors.
4. **Timing Inversion in Stage 2 Matching:** In the legacy sequence, staffing matching was executed at Level 0 based on expected continuous state shares $y_{s,j}$, before Level 2 determined the actual physical boundaries. Because sales books $S_i(z)$ are clustered discrete point masses rather than continuous fluids, matching on continuous state shares produced substantial book displacement once realized boundaries were finalized.

3 The Support-Based Exact Reformulation Framework

To resolve the failure modes of the legacy pipeline, we developed and deployed the **Support-Based Exact Reformulation Framework**. This formulation replaces both the flow-based Level 0 MILP and the heuristic Level 2 repair with an exact, support-enumerated mathematical program.

3.1 Mathematical Formulation

Let $G = (V, E)$ denote the planning graph (where V represents states or sub-state macro units such as Northern California and Southern California, with $|V| = 50$).

Definition 1 (Admissible Support Space). A subset of vertices $S \subseteq V$ is an **admissible support** ($S \in \mathcal{S}$) if and only if it satisfies:

1. **Induced Subgraph Contiguity:** The induced subgraph $G[S] = (S, E[S])$ is connected.
2. **Cardinality Bound:** $|S| \leq \bar{s}$, where $\bar{s} \in \{5, 6\}$ limits administrative sprawl.
3. **Spatial Diameter Ceiling:** The maximum pairwise geographic centroid distance satisfies:

$$\max_{u,v \in S} \|c_u - c_v\|_2 \leq D_{\max} \quad (17)$$

where $D_{\max} = 900$ km for standard CONUS supports, with an expanded ceiling of 1,200 km for sparse Pacific Northwest supports (e.g., WA, OR, ID).

By pre-computing $\mathcal{S}$, contiguity is enforced **a priori by construction**. Graph cycle checks and flow variables are completely eliminated from the optimization problem.

3.2 The Master Support Partitioning MILP

Let K denote the exact required number of districts. For each admissible support $S \in \mathcal{S}$, let $x_S \in \{0, 1\}$ denote the binary support activation indicator, and for each unit $v \in S$, let $y_{v,S} \in [0, 1]$ denote the continuous share of unit v allocated to district support S .

The Master Problem is formulated as:

$$\min_{x,y} \sum_{S \in \mathcal{S}} \sum_{v \in S} \text{Cost}(v, S) y_{v,S} + \rho \sum_{S \in \mathcal{S}} \text{Diam}(S) x_S \quad (18)$$

$$\text{s.t.} \quad \sum_{S \in \mathcal{S}} x_S = K \quad (19)$$

$$\sum_{S \in \mathcal{S}: v \in S} y_{v,S} = 1 \quad \forall v \in V \quad (20)$$

$$\eta x_S \leq y_{v,S} \leq x_S \quad \forall S \in \mathcal{S}, v \in S \quad (21)$$

$$L_K x_S \leq \sum_{v \in S} M_v y_{v,S} \leq U_K x_S \quad \forall S \in \mathcal{S} \quad (22)$$

$$y_{v,S} \in \{0, 1\} \quad \forall v \in V_{\text{indivisible}}, S \in \mathcal{S} \quad (23)$$

$$\sum_{S \in \mathcal{S}: \{u,v\} \subseteq S} x_S \geq 1 \quad \forall (u, v) \in \mathcal{C}_{\text{couple}} \quad (24)$$

$$x_S \in \{0, 1\}, \quad y_{v,S} \geq 0 \quad \forall S \in \mathcal{S}, v \in S \quad (25)$$

where:

- Constraint (19) enforces the exact district count K .
- Constraint (20) enforces 100% complete, non-overlapping coverage across all units, guaranteeing **zero unheld opportunity mass**.
- Constraint (21) guarantees that active contacts receive at least share $\eta = 0.05$ (preventing negligible slivers).
- Constraint (22) enforces hard capacity bounds $[L_K, U_K] = [\tau_K(1 - \epsilon), \tau_K(1 + \epsilon)]$ with $\epsilon \in [0.10, 0.12]$.
- Constraint (23) enforces that smaller states ($V_{\text{indivisible}}$) remain intact as whole units without boundary splitting.
- Constraint (24) enforces macro-regional pairing (such as CA1 and CA2 coupling restrictions).

3.3 Theoretical Guarantees of the Support Formulation

Theorem 1 (Contiguity and Conservation Guarantee). *Every feasible solution (x^*, y^*) to the Master Support Partitioning MILP (18)–(25) induces a partition of V into exactly K districts satisfying:*

1. **100% Rook Contiguity:** *Every activated district S with $x_S^* = 1$ is an induced connected subgraph of G .*
2. **Zero Unheld Mass:** $\sum_{S: x_S^*=1} \sum_{v \in S} M_v y_{v,S}^* = \sum_{v \in V} M_v$ exactly, with 0.00% unheld mass.
3. **Zero Heuristic Relaxation Slack:** *All district masses strictly satisfy $L_K \leq M(S) \leq U_K$ without post-hoc σ -tolerance expansions.*

Proof. By Definition 1, the enumeration procedure admits only subsets S whose induced subgraph $G[S]$ is connected. Because $x_S^* \in \{0, 1\}$ and $\sum_S x_S^* = K$, exactly K subsets are activated, each inherently connected. Summing (20) over all $v \in V$ weighted by M_v yields:

$$\sum_{v \in V} M_v = \sum_{v \in V} M_v \sum_{S: v \in S} y_{v,S}^* = \sum_{S \in \mathcal{S}} \sum_{v \in S} M_v y_{v,S}^* = \sum_{S: x_S^*=1} M(S)$$

Thus, total system mass is preserved exactly. Finally, constraint (22) binds the continuous variables directly, ensuring that no district can violate capacity bounds. $\square$

3.4 Computational Complexity and Independent Certification

Because the support space $|\mathcal{S}|$ on G for $\bar{s} \leq 6$ and $D_{\max} \leq 900$ km contains only 1,200–4,500 valid candidates, the Master MILP comprises fewer than 5,000 binary columns and fewer than 300 constraints. Solved using the HiGHS branch-and-cut engine, the problem achieves **provable global optimality in 0.8–2.8 seconds**, completely bypassing the combinatorial timeouts of the legacy flow model.

To ensure verification integrity, every generated solution is certified by an **independent, solver-agnostic audit script** (`tools/group2_support.py`) verifying:

- Exact district count ($= K$).
- Full mass coverage ($\sum_S y_{v,S} = 1.000000 \pm 10^{-6}$).
- Strict breadth-first-search (BFS) rook contiguity on G .
- Strict pairwise centroid distance $\leq D_{\max}$.
- Contact share compliance ($y_{v,S} \geq 0.05$).
- Absolute capacity band compliance ($M(S) \in [L, U]$).
- Macro purity and split-cap compliance.

4 Unified Multi-Channel Architecture and the Western Merged Realization

4.1 The Sparse Geography Dilemma

A critical operational challenge arises in the Western United States across seven rural states:

$$S_{\text{west}} = \{\text{CO, ID, MT, ND, NE, SD, WY}\} \quad (26)$$

These states encompass vast territorial expanses with low commercial sales opportunity:

- Total descaled opportunity in Wealth (WH) and Financial Institutions (FI): $M(S_{\text{west}}) = 431.26$.
- Total descaled opportunity across all channels (National + WH + FI): $M(S_{\text{west}}) = 689.01$.

Under independent single-channel planning, dedicating dedicated National representatives to this footprint is mathematically infeasible: the National target district size is $\tau_N \approx 615$, requiring combining non-contiguous population centers across multiple time zones and violating the 900 km centroid diameter ceiling.

4.2 The Dedicated Regional Merged Channel (WIFI/WHFI)

To resolve this structural tension without compromising single-channel focus in high-density markets, the unified architecture introduces a dedicated regional merged channel:

$$K = N + WH + FI + WIFI \quad (27)$$

where:

- $N \in \{12, 13, 14\}$: Standalone National districts across the remaining 42 states.
- $WH \in \{10, 11, 13, 14\}$: Standalone Wealth districts.

- $FI \in \{21, \dots, 26\}$: Standalone Financial Institutions districts.
- $WIFI \in \{0, 1, 2, 3, 4, 5, 6\}$: Dedicated merged districts covering S_{west} , in which representatives service both Wealth and FI (or National, Wealth, and FI) accounts within compact, drivable geographic clusters.

4.3 The 36-Scenario Pareto Catalog

Using the Support-Based Exact Reformulation, we generated and certified a master catalog of **36 Pareto-optimal operational scenarios** spanning total districts $K \in \{46, \dots, 53\}$:

- **Recommended Baseline** ($K = 51$): `51_total_13n_11wh_24fi_3wifi` (13 National, 11 WH, 24 FI, 3 WIFI). Achieves 100% contiguity, 0.00% unheld mass, and ideal balance.
- **Varying WIFI Exploration** ($K \in \{48, 49, 50\}$): Holding $FI = 21$ and $WH = 11$ strictly fixed while stepping $WIFI \in \{2, 3, 4, 5, 6\}$ across National tiers $N \in \{14, 13, 12\}$:
 - Tier $N = 14$ ($WIFI \in \{2, 3, 4\}$):
`48_total_14n_11wh_21fi_2wifi`, `49_total_14n_11wh_21fi_3wifi`,
`50_total_14n_11wh_21fi_4wifi`
 - Tier $N = 13$ ($WIFI \in \{3, 4, 5\}$):
`48_total_13n_11wh_21fi_3wifi`, `49_total_13n_11wh_21fi_4wifi`,
`50_total_13n_11wh_21fi_5wifi`
 - Tier $N = 12$ ($WIFI \in \{4, 5, 6\}$):
`48_total_12n_11wh_21fi_4wifi`, `49_total_12n_11wh_21fi_5wifi`,
`50_total_12n_11wh_21fi_6wifi`
- **Multi-Tier Expansion** ($K \in \{51, 52, 53\}$): Systematically trading off National specialization ($N = 12, 13, 14$) against FI granularity ($FI = 24, 25, 26$) and WH capacity ($WH = 10, 11, 13, 14$).

All 36 scenarios are compiled into `summary.csv` and `scenarios.csv` (1,166,040 cell assignment records), certified with 0 contiguity violations and 0.00% unheld mass.

5 Data Descaling and Scale Invariance

5.1 The Descaled Data Representation

As implemented in `td/instance.py`, the raw instance data undergoes a canonical descaling transformation governed by $\kappa > 0$, defined as the median positive ZIP opportunity across the footprint:

$$m_{\text{rel}}(z) = \frac{M_z}{\kappa}, \quad s_i(z) = \frac{S_i(z)}{M_z} \quad (28)$$

Upon loading, the pipeline reconstructs:

$$M(z) = m_{\text{rel}}(z) = \frac{\text{real } M_z}{\kappa}, \quad S_i(z) = s_i(z) \cdot m_{\text{rel}}(z) = \frac{\text{real } S_i(z)}{\kappa} \quad (29)$$

Under this transformation, the national opportunity sum across CONUS is $W_0 \approx 3,268.41$ descaled units, and district target capacity becomes $\tau \approx 180\text{--}200$ descaled units (rather than $\approx \$10^9$ raw dollars).

5.2 Proposition: Exact Scale Invariance of Nash Welfare

Proposition 1 (Scale Invariance of Nash Social Welfare). *Let $\mathcal{I} = (G, (S_i)_{i=1}^n, S_{\text{free}}, M)$ be an instance. Scaling every currency field uniformly by $\kappa > 0$ yields instance $\kappa\mathcal{I}$. For every district assignment w and any perimeter penalty weight $\rho \geq 0$:*

1. $u_i^{\kappa\mathcal{I}}(z) = \kappa u_i^{\mathcal{I}}(z)$ and $g_{i,j}^{\kappa\mathcal{I}} = \kappa g_{i,j}^{\mathcal{I}}$.
2. $\sum_{i=1}^n \log g_{i,\pi(i)}^{\kappa\mathcal{I}} = \sum_{i=1}^n \log g_{i,\pi(i)}^{\mathcal{I}} + n \log \kappa$.
3. *The additive shift $n \log \kappa$ is strictly independent of the allocation w and assignment π . Consequently:*

$$\arg \max_{w,\pi} \left(\sum_{i=1}^n \log g_{i,\pi(i)}^{\kappa\mathcal{I}} - \rho P(w) \right) \equiv \arg \max_{w,\pi} \left(\sum_{i=1}^n \log g_{i,\pi(i)}^{\mathcal{I}} - \rho P(w) \right) \quad (30)$$

All objective differences, certificates, and optimality gaps are identical across $\mathcal{I}$ and $\kappa\mathcal{I}$.

5.3 Why Multiplicative Normalization Fails to Restore Curvature

A natural question is whether normalizing the utility function by overall opportunity (either instance-wide M_{tot} or district-level M_j) solves the flat-curvature issue of Nash welfare. Mathematically, it cannot:

1. **Global Normalization** ($g_{i,j}/M_{\text{tot}}$): By Proposition 1, dividing by a global constant simply subtracts $n \log M_{\text{tot}}$ from the objective. The argmax, gradients, and Hessian structure are completely untouched.
2. **District Normalization** ($g_{i,j}/M_j$): Expanding $g_{i,j}$:

$$\frac{g_{i,j}}{M_j} = \underbrace{(c_1 - c_2) \frac{S_i(V_j)}{M_j}}_{\Delta_{i,j}/M_j} + \underbrace{\left[c_2 \frac{T(V_j)}{M_j} + c_{\text{free}} \frac{S_{\text{free}}(V_j)}{M_j} + \lambda \right]}_{G_{0,j}/M_j} \quad (31)$$

Dividing by M_j scales numerator and denominator proportionally. The critical ratio:

$$\frac{\Delta_{i,j}/M_j}{G_{0,j}/M_j} = \frac{\Delta_{i,j}}{G_{0,j}} \quad (32)$$

is strictly **dimensionless**. Because $G_{0,j} \gg \Delta_{i,j}$ (ambient district opportunity and cross-selling dwarf individual book differences), the Taylor expansion remains linear:

$$\log \left(\frac{G_{0,j}}{M_j} + \frac{\Delta_{i,j}}{M_j} \right) \approx \log \left(\frac{G_{0,j}}{M_j} \right) + \frac{\Delta_{i,j}}{G_{0,j}} \quad (33)$$

Multiplicative scaling cannot alter the dimensionless ratio Δ/G_0 . Only an **additive translation** ($g - d$) can shift the operating point of $\log(x)$ toward its steep curvature regime.

6 The Disagreement Point Dilemma: Classical EF1 vs. Feasible Envy-Freeness

6.1 The Zero Threat Point and the Caragiannis Guarantee

In classical cooperative bargaining theory (Nash, 1950), the Nash product is maximized over gains above a disagreement point $d = (d_1, \dots, d_n)$:

$$\max_{\pi} \prod_{i=1}^n (u_i(\pi_i) - d_i) \quad (34)$$

In `docs/MODEL_FULL.md` and `docs/math_note/math_note.tex`, the baseline is fixed at $d = \mathbf{0}$. The foundational justification rests on the landmark theorem of Caragiannis et al. (2019):

Theorem 2 (Caragiannis et al., 2019). *For unconstrained fair division of indivisible goods under additive valuations and a zero disagreement point $d = \mathbf{0}$, any allocation maximizing the Nash Social Welfare $\prod_{i=1}^n u_i(A_i)$ is simultaneously:*

1. **Pareto Optimal (PO)**, and

2. **Envy-Free up to One Item (EF1)**: *For any pair of agents i, k , there exists an item $z \in A_k$ such that:*

$$u_i(A_i) \geq u_i(A_k \setminus \{z\}) \quad (35)$$

6.2 Why Shifting to $d > 0$ Voids Classical EF1

The proof of Theorem 2 relies directly on local exchange ratios: if agent i envies agent k by more than one good, shifting item $z \in A_k$ to A_i strictly increases the Nash product:

$$\left(1 + \frac{u_i(z)}{u_i(A_i)}\right) \left(1 - \frac{u_k(z)}{u_k(A_k)}\right) > 1 \quad (36)$$

When an additive disagreement vector $d > \mathbf{0}$ is introduced:

$$\left(1 + \frac{u_i(z)}{u_i(A_i) - d_i}\right) \left(1 - \frac{u_k(z)}{u_k(A_k) - d_k}\right) \quad (37)$$

If rep i is driven near their reservation utility ($u_i - d_i \rightarrow \epsilon$), the marginal ratio $\frac{u_i(z)}{\epsilon}$ diverges to infinity. The optimizer responds by aggressively transferring goods from unconstrained agents to agent i , even when agent i already possesses substantial bundle value. As established in the literature ledger (`docs/foundations/LIT_economic-theory.md`, row A4), subtracting a constant strictly voids the Caragiannis guarantee; no deterministic indivisible EF1 analogue exists for $d > 0$.

6.3 The Reality: Capacity Bands Already Invalidate Classical EF1

Crucially, `docs/foundations/DOMAIN_economic-theory.md` (§2.1, §2.8) proves that the territory design problem **already violates the hypotheses of Theorem 2**:

1. **Quantity Capacity Bands**: Districts must satisfy $L \leq M(V_j) \leq U$. The optimizer cannot execute the arbitrary item transfers required by the Caragiannis exchange step without violating the capacity band.
2. **Topological Contiguity**: Items (ZIPs) cannot be transferred if doing so disconnects the territory graph.

Empirical measurement on the CONUS footprint demonstrates a representative utility spread of over 60% (`MODEL_U8-band`). Thus, classical unconstrained EF1 is already defunct on the ground.

6.4 Transition to Feasible Envy-Freeness (FEF_x)

Under Generalized Assignment Constraints, Barman et al. (2023) establish the appropriate axiomatic fairness concept:

Definition 2 (Feasible Envy-Freeness up to Any Item (FEFx)). Let $\mathcal{F}_i$ denote the set of all feasible territories for representative i (satisfying drivability from home base, spatial contiguity, and capacity bands $[L, U]$). An allocation A is **Feasible Envy-Free up to Any Item (FEFx)** if for every pair of agents i, k :

$$A_k \in \mathcal{F}_i \implies \forall z \in A_k, u_i(A_i) \geq u_i(A_k \setminus \{z\}) \quad (38)$$

If territory A_k is not feasible for representative i ($A_k \notin \mathcal{F}_i$), agent i cannot claim legitimate envy against A_k .

As succinctly formulated in the economics charter: “*A wholesaler cannot envy a bundle they could not work.*” Under FEFx, cross-regional comparisons (e.g., an Ohio representative envying a Florida territory) are pruned by operational feasibility.

6.5 Dropping Anonymity: Asymmetric Entitlements and Claims

In sales force restructuring, representatives are not anonymous participants with equal claims to unowned goods. They hold **asymmetric historical entitlements** defined by their legacy books $S_i(Z)$.

We formulate the staffing problem as an **asymmetric claims problem** (O’Neill, 1982; Thomson, 2003). The objective becomes:

$$\max_{\pi} \sum_{i \in R} \log(g_{i, \pi(i)} - d_i) \quad (39)$$

subject to the **Bargaining Set Non-Emptiness Guard**:

$$d_i = \alpha_i S_i(Z) + d_{\text{floor}} \quad (40)$$

where $d_{\text{floor}} = \gamma \min_j G_{0,j}$ ($\gamma \in [0.5, 0.8]$) ensures a common safety net, and $\alpha_i \in (0, 1)$ is determined via proportional claims bankruptcy scaling:

$$\alpha_i = \min \left(1.0, \frac{M(\text{Region})}{\sum_{k \in \text{overlap}} S_k(\text{Region})} \right) \cdot (1 - \epsilon) \quad (41)$$

This formulation achieves three vital properties:

1. **Guaranteed Feasibility:** $g_{i,j} - d_i > 0$ for all valid district assignments.
2. **Restored Curvature:** Denominators drop from ≈ 200 to $\approx 10\text{--}20$, amplifying marginal resistance to book starvation by over $100\times$.
3. **FEFx Compliance:** Envy is bounded with respect to operationally feasible assignments.

7 Planar Polygon Dissolution and Full Geographic Integrity

A major downstream challenge in territorial optimization is translating discrete cell assignments into continuous, boundary-complete geometries for executive dashboards and geographic information systems (GIS).

7.1 Eliminating the Swiss-Cheese Effect

In naive GIS exports, plotting only commercial opportunity ZIP codes (3,713 populated ZIPs out of 33,791 Census ZCTAs) creates vast white voids in rural areas. Furthermore, standard administrative boundaries fail when states like California or Texas are split across multiple districts.

To solve this, we implemented the **Planar Polygon Dissolution Pipeline** (see scripts `geom_export.py` and `export_tableau_datasets.py` in `tools/`):

1. **Voronoi Proximity Tessellation:** Unpopulated and rural ZIP centroids are assigned to the nearest commercial district center in Lambert Azimuthal Equal Area (LAEA) projection.
2. **State Boundary Clipping:** The dissolved cell polygons are strictly intersected with official US Census state boundary shapefiles.
3. **WGS84 Transformation:** Cleaned geometries are transformed to standard coordinates (EPSG:4326) and exported to `tableau_district_reach.geojson` (1,809 district polygons across all 36 scenarios).

7.2 The Master 33,791-ZCTA Crosswalk

For relational database ingestion, we constructed the full nationwide crosswalk:

$$D = 33,791 \text{ Census ZCTAs} \times 5 \text{ sub-channels} \times 36 \text{ scenarios} = 6,082,380 \text{ rows} \quad (42)$$

compressed into `all_scenarios_nationwide_zcta_long.zip` (27.2 MB) under the 12-column canonical relational schema:

$$\mathcal{S}_{\text{schema}} = (\text{scenario}, \text{zip_code}, \text{current_channel}, \text{canonical_channel}, \\ \text{state}, \text{model_channel}, \text{bundle}, \text{district}, \\ \text{district_channels}, \text{rep}, \text{m_rel}, \text{has_commercial_opportunity})$$

This bridges mathematical district definitions directly with ERP and CRM reporting without spatial omission.

8 Synthesis of Mathematical Paradigms

9 Conclusion and Operational Status

The evolution from the legacy three-level heuristic decomposition to the **Support-Based Exact Reformulation** resolves the core mathematical and computational bottlenecks in sales territory design:

1. **Computational Scalability:** Replacing combinatorial flow variables with pre-enumerated connected supports shrinks solve times from > 600 seconds to under 3 seconds.
2. **Mathematical Rigor:** District contiguity, exact capacity bands, and zero unheld opportunity mass are guaranteed by construction, verified by an independent certificate validator.
3. **Geographic Realism:** Dedicated regional merged channels (WIFI) cleanly solve sparse rural Western territories while preserving channel specialization nationwide.
4. **Axiomatic Fairness:** Shifting from unconstrained EF1 to Feasible Envy-Freeness (FEF_x) under asymmetric claims restores concavity to the welfare function and protects incumbent sales books.

All 36 scenario configurations, full-scale GIS polygons, and relational database exports are certified, documented, and published in the repository.

Component	Mathematical Paradigm	Theoretical Strengths	Practical Performance / Status
Legacy Level 0	Multi-Commodity Flow MILP	Strict Pareto hierarchy; compact flow trees.	Combinatorial timeout on multi-channel problems ($> 600s$).
Legacy Level 2	Optimal Transport + Graph Repair	Optimal Euclidean compactness via power cells.	Induces disconnected islands; $\pm 2\%$ capacity slacks.
Latest Approach	Support-Based Exact MILP	100% rook contiguity by construction ; 0.00% unheld mass; exact capacity bands.	Solves in < 3 seconds via HiGHS; certified across all 36 scenarios.
Western Model	Dedicated Merged WIFI	Resolves sparse density without diluting urban channels ($K = N + WH + FI + WIFI$).	Successfully realized for $WIFI \in [1, 6]$ across $K \in [46, 53]$.
Stage 2 Staffing	Asymmetric Claims Matching	Operates $\log(g_i - d_i)$ in steep curvature regime; protects legacy books.	Scale invariant; polynomial solve via Kuhn-Munkres.
Fairness Audit	FEFx Matrix Audit	Evaluates envy only over operational feasible set $\mathcal{F}_i$.	Replaces classical unconstrained EF1, which is voided by capacity bands.
Spatial GIS	Planar Polygon Dis-solution	Exact state splits; zero Swiss-cheese omissions in Tableau.	1,809 polygons in GeoJSON; 6.08M ZCTA records.

Table 1: Comprehensive Comparison: Legacy Decomposition vs. Latest Production Architecture.